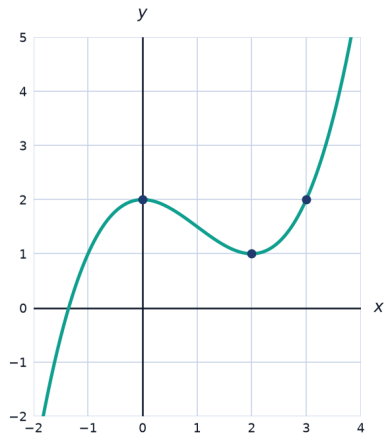
Integer solutions to linear programming applications (Extended)

Core Concepts: Integer solutions to linear programming applications (Extended)

1 For the model $f(x) = 5x + 3$ in integer solutions to linear programming applications (extended), evaluate $f(2)$.



a. 13

b. 18

c. 10

d. 17

2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

3 A data set recorded for integer solutions to linear programming applications (extended) consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

Procedural Fluency & Algebraic Manipulation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 For the model $f(x) = 5x + 3$ in integer solutions to linear programming applications (extended), evaluate $f(3)$.

a. 18

b. 23

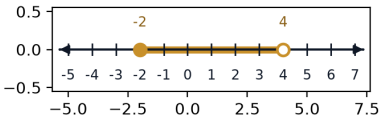
c. 15

d. 22

5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$



c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

6 A data set recorded for integer solutions to linear programming applications (extended) consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

Graphical & Geometric Interpretation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 For the model $f(x) = 5x + 3$ in integer solutions to linear programming applications (extended), evaluate $f(4)$.

a. 23

b. 28

c. 20

d. 27

8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

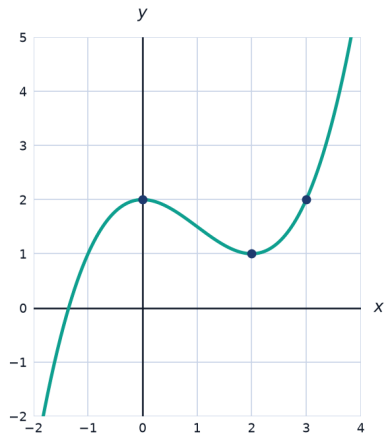
9 A data set recorded for integer solutions to linear programming applications (extended) consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5



Exam-Style Problem Solving & Modeling

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

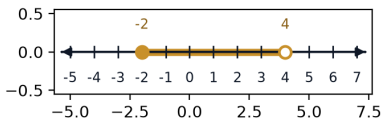
10 For the model $f(x) = 5x + 3$ in integer solutions to linear programming applications (extended), evaluate $f(5)$.

a. 28

b. 33

c. 25

d. 32



11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

12 A data set recorded for integer solutions to linear programming applications (extended) consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

© ClipSAT: By Mr. Mohamed Abdallah +966597688647